

Practices

Practice 1 - Ressourcen

Prerequisites

You have checked out “lessons/01-initial-vaadin-application”.

Motivation

Become familiar with the Vaadin Projektstructure.

Time

5 Minutes

Task

Add a picture in a `Div` to the class `MainView`.

Bonus

Center the picture.

Practice 2 - Layers

Prerequisites

Practice 1 is completed or you checked out branch “lessons/01-initial-vaadin-application”.

Motivation

Learn splitting an application according to the software layers.
Practice MVP.

Alternative

Check out branch “lessons/02-multi-module-vaadin-application” and understand how classes had to move.

Time

120 Minutes

Task 1

Split the application in multiple modules, which reflect the layers Persistenz, Service, Presenter and UI.

Hint: The architecture tests will not function anymore, since they were based on package checks which followed a different scheme.

Task 2

Create an own module, which abstracts the access to the presenter layer via interfaces. This enables you to easily exchange the persistence layer with other persistence implementations.

Bonus 1

Create a persistence module, which implements JDBC, and use it instead of the JPA implementation.

Practice 3 - Spring Security

Prerequisites

Practice 2 is completed or branch “lessons/02-multi-module-vaadin-application” is checked out.

Motivation

Spring Security is a key element for secured Applications.
Spring Security is required for the next step.

Time

30 Minutes

Task

Add the dependency `spring-boot-starter-security` to the `pom.xml` in the root module.

Create a Spring Configuration class in the UI module.

Create a helper class `SecurityUtils` having the method `getUser()` in the service module.

Create a login view.

Adapt the user menu in the class `MainLayout` and add the method `logout()`.

Add the necessary access Annotations to all views and layouts.

As support you can use the tutorial here

<https://vaadin.com/docs/latest/flow/security/enabling-security>. Different classes can be copied from this tutorial.

Bonus

Move all code elements to the right module.

Practice 4 - Asynchronous Processes

Prerequisites

Practice 3 is completed or branch "lessons/03-spring-security" is checked out.

Motivation

Practicing asynchronous processes.

Time

60 Minutes

Task 1

Create a helper class in the service layer, containing 2 methods to apply the SecurityContext to another thread and cleans it up after the task has been executed. One Method should take in a Runnable and the other a Supplier as a long running task and execute them.

Task 2

Create a helper class in the UI Layer which eases the access to the UI Thread.

Task 3

Create a progress dialog as defined in the presentation (/presentations/en/04 Asynchronous Processes.pdf)

Task 4

Add the @Push annotation to the Application class.

Enhance the process of adding a Task (TaskListPresenter#createTaskClicked(...)). Add an artificial long running task after the call of the taskService method and execute it in a thread. Please make sure that the following 3 view calls are executed after this long running task and the progress dialog is shown while the long running task is executed.